

RITIKA SHRESTHA

SOFTWARE DEVELOPMENT | DATA SCIENCE AND INTELLIGENT SYSTEMS

Contact: +33 7 83 48 41 33 · ritika.shrestha@telecom-sudparis.eu
Socials: ritika-shrestha.com.np/ · linkedin.com/in/ritikashrestha/ · github.com/RitikaShrestha-me
Availability: 6-month end-of-study internship from February 2027 · International student (VLS-TS)
Location: Flexible to relocate · Télécom SudParis can sign the convention de stage
Languages: English (C1) · Nepali (Native) · Hindi (Fluent) · French (Beginner, A1)

CAREER PROFILE

Dual Degree Master's student in Data Science and Network Intelligence at Télécom SudParis, with 2+ years of software development experience building data-driven applications. Skilled in front and backend services, data pipelines, and machine learning, with a track record of improving product performance and delivering dashboards that turn data into clear insights. Works in Agile teams to translate requirements into practical, reliable solutions. Combines analytical thinking with a builder mindset to solve real-world problems using data and technology.

EDUCATION

Institut Polytechnique de Paris - Télécom SudParis, France 2026 – present
MSc. in Data Science and Network Intelligence

Glacial Lake Outburst Flood (GLOF) Early Warning System | IoT & Digital Twins
- Developed a data pipeline integrating simulated sensor streams, validation, and automated risk scoring.
- Defined system requirements with the team, designed monitoring dashboards, and tested five operational scenarios.
- Combined hazard and population-exposure indicators to classify risk and trigger alerts.

Asian Institute of Technology, Thailand 2025 – present
M.Eng in IoT Systems Engineering | First year: Grade A (3.94/4) | HM King's Scholarship recipient

IoT Ranger | IoT Systems Project
- Built a monitoring platform that collects real-time device telemetry and maintains historical data for analysis.
- Developed dashboards, alerts, and monitoring services to improve visibility into device performance and system health.
- Designed modular backend services and data flows to support reliable, scalable monitoring.
- Containerized the platform with Docker and documented the architecture and data flows.

AI-Assisted Wireless User Scheduling | Machine Learning
- Generated a 20,000-sample dataset and trained a Random Forest model (92.95% test accuracy), benchmarked against three scheduling algorithms and within 0.5% of Proportional Fair throughput.

Visvesvaraya Technological University (VTU), India 2018 – 2022
B.E. in Computer Science & Engineering | 8.57/10 CGPA

DOODLEO | AI-Powered Wireframe-to-Code Generation
Bachelor's Final-Year Project | Python, CNN, LSTM, OpenCV, HTML
- Developed an AI-based approach to convert hand-drawn webpage wireframes into HTML code.
- Designed a synthetic dataset pipeline (layouts, sketches, HTML via a domain-specific language) and explored a CNN encoder with an LSTM decoder.

WORK EXPERIENCE

Viveka Services – Associate Frontend Developer Nov 2023 – Feb 2025
- Collaborated with product, backend, and QA teams to deliver features for a health insurance platform serving 450,000+ users.
- Translated product requirements into Jira tickets and user stories and contributed to Agile sprint delivery.
- Improved application performance by reducing page load times by 30% and bundle size by approximately 25%.
- Built reusable components and supported scalable, maintainable product development.

BalooChat Technology – Software Development Engineer Sept 2022 – Apr 2023
- Developed mobile application features and integrated REST APIs in collaboration with backend and design teams.
- Improved application maintainability and performance through reusable components and efficient caching.

TaxAdda Private Limited – React Frontend Developer June 2021 – Dec 2021
- Built a seller dashboard with authentication and transaction-processing features for an e-commerce tax-compliance platform.
- Developed tools for filtering, processing, and exporting marketplace transaction data.

TECHNICAL SKILLS

- **Web development:** JavaScript, TypeScript, React, REST APIs, WebSockets, reusable components, performance optimization.
- **Backend & data:** Python, FastAPI, MongoDB, InfluxDB, SQL, MQTT, data pipelines, time-series data processing.
- **Systems & tooling:** Mac, Linux, Windows, Ubuntu, Docker, Git
- **Monitoring & visualization:** Grafana, dashboard development, alerting, performance benchmarking.
- **Machine learning:** scikit-learn, Random Forest, LightGBM, CatBoost, Pandas, NumPy, model evaluation, cross-validation.
- **Ways of working:** Agile/Scrum, Jira, requirements analysis, user stories, technical documentation, cross-functional collaboration.

Additional projects and work experience are described in detail on my Website profile: ***ritika-shrestha.com.np/***